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Factors Associated with High Blood Cholesterol among Indigenous Adults in Canada: A Cross-sectional Analysis of the 2022 Canadian Community Health Survey

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Authors' contributions

This work was carried out in collaboration among all authors. All authors read and approved the final manuscript.

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Abstract

Background: High blood cholesterol is a major modifiable risk factor for cardiovascular disease, but evidence on associated factors among Indigenous adults in Canada remains limited.
Objective: To identify demographic, socioeconomic, lifestyle, and clinical factors associated with high blood cholesterol among Indigenous adults in Canada.

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Methods: A cross-sectional study was conducted using data from the 2022 Canadian Community Health Survey. The analysis included 1,632 Indigenous adults aged ≥18 years, representing 607,893 adults after survey weighting. Survey-weighted multivariable logistic regression with bootstrap replicate weights estimated adjusted odds ratios (aORs) and 95% confidence intervals (CIs) for self-reported high blood cholesterol.

Results: Older age, diabetes, hypertension, and province of residence were independently associated with high blood cholesterol. Compared with adults aged 18–34 years, higher odds were observed among those aged 50–64 years (aOR=5.16, 95% CI 1.88–14.16) and ≥65 years (aOR=5.80, 95% CI 2.05–16.42). Diabetes (aOR=7.06, 95% CI 2.77–17.99) and hypertension (aOR=2.36, 95% CI 1.18–4.70) were also associated with higher odds. Compared with Atlantic Canada, lower odds were observed in Quebec (aOR=0.25, 95% CI 0.07–0.98) and the Prairies (aOR=0.28, 95% CI 0.10–0.81). Sex, education, body mass index, smoking, and alcohol consumption were not significantly associated with the outcome.

Conclusions: Older age, diabetes, hypertension, and geographic region were associated with high blood cholesterol among Indigenous adults in Canada. These findings support integrated cardiovascular risk assessment and early identification of dyslipidaemia in Indigenous populations.

Keywords: Indigenous health; high blood cholesterol; dyslipidemia; cardiovascular disease; canadian community health survey.

1. Introduction

High blood cholesterol is a major modifiable risk factor for cardiovascular disease and significantly contributes to the development of atherosclerosis, coronary heart disease, stroke, and premature death (Goldberg et al., 2019). Heart disease continues to be one of the most common causes of both illness and death, with many scholars arguing that the identification and management of high blood cholesterol are critical elements in the prevention of cardiovascular disease (Packer et al., 2020; Vervoort et al., 2022). High blood cholesterol frequently occurs alongside other cardiometabolic conditions such as obesity, hypertension, and diabetes, increasing the likelihood of adverse health outcomes (Ray et al., 2017). Therefore, there is a need to identify, understand, and address the various factors contributing to high blood cholesterol when developing appropriate public health interventions targeting high blood cholesterol and preventing cardiovascular disease in Canada (Ray et al., 2019). A review of lipid-related cardiovascular mechanisms similarly describes dyslipidaemia as an important cardiovascular risk factor (Rai, 2021). Contemporary clinical guidance continues to situate dyslipidaemia evaluation within comprehensive cardiovascular risk assessment (Saeed et al., 2026).

Many Indigenous peoples, including First Nations, Métis, and Inuit Peoples, face higher rates of cardiovascular disease and risk factors than the general non-Indigenous population in Canada (Tobe et al., 2019). A variety of historical, systemic, and current factors influence these health differences between Indigenous Peoples and other Canadians, including colonisation, systemic racism, and economic disadvantage (Tobe et al., 2019). National surveillance data on chronic disease outcomes in Indigenous Peoples have demonstrated that disparities in chronic disease outcomes in these populations persist (Umaefulam et al., 2022). These findings highlight the importance of addressing modifiable cardiovascular disease risk factors among Indigenous populations (McCallum et al., 2023). A recent review that included Canadian Indigenous populations likewise described persistent cardiovascular inequities and important limitations in population-level data (Brown et al., 2026).

Moreover, as adults age, their risk of developing high blood cholesterol increases; however, an individual's sex, educational attainment, household income, employment status, and geographic location have been shown to account for variations in cholesterol prevalence and cardiovascular risk (Mackinnon et al., 2022; Zhao et al., 2024). Cigarette smoking, alcohol consumption, physical inactivity, poor dietary habits, and obesity can contribute to the development of dyslipidaemia (Chai et al., 2024; Pearson et al., 2021). In addition, conditions such as hypertension, diabetes, and cardiovascular disease (CVD) occur comorbidly with elevated cholesterol because they share common metabolic and behavioural risk factors with dyslipidaemia (Hu et al., 2022). Understanding how these demographic, socioeconomic, and lifestyle/climatic factors influence cholesterol status among Indigenous adults is critical for identifying those at highest risk of developing CVD and determining how to implement prevention measures (Hassen et al., 2022). Recent epidemiological literature also considers diabetes, hypertension, and cardiovascular disease as interconnected chronic conditions in population-health assessment (David et al., 2024).

Although an increasing amount of information is available on CVD risk factors in Canadian populations, very few studies using representative data have specifically assessed high blood cholesterol among Indigenous adults (McCallum et al., 2023). The current literature focuses on the broader Canadian population; even studies that have assessed Indigenous adults have tended to focus on different regions rather than evaluating the specific population of Indigenous persons (Roy et al., 2024). The emphasis has been on broader cardiovascular disease variables rather than on simultaneously assessing the interaction among demographic, socioeconomic, lifestyle, and clinical variables contributing to high blood cholesterol in Indigenous persons (Roy et al., 2024). Recent systematic reviews of hypercholesterolaemia have demonstrated a substantial burden of illness within the Indigenous population; however, there remains limited information from contemporary epidemiological studies using national datasets (Kelly et al., 2019).

The Canadian Community Health Survey (CCHS) is a countrywide random-sample survey developed by Statistics Canada that collects comprehensive data on Canadians' health, chronic disease states, health care use, and other elements of the social determinants of health (Riediger et al., 2010). The CCHS provides information on demographic and socioeconomic characteristics, health behaviours, and chronic health conditions, making it a suitable data source for examining factors associated with high blood cholesterol among Indigenous adults represented within the survey sampling frame (Statistics Canada, 2025). Due to its countrywide representative sample, standardised methods of data collection, and ability to provide information on population-level disparities in cross-sectional analyses, the CCHS provides researchers and decision-makers with the evidence needed to make informed public health decisions (Shahunja et al., 2024).

Despite the availability of national survey data, contemporary evidence remains limited on the combined demographic, socioeconomic, lifestyle, and clinical factors associated specifically with high blood cholesterol among Indigenous adults in Canada.

1.1 Objective

The objective of this study was to identify the demographic, socioeconomic, lifestyle, and clinical factors associated with high blood cholesterol in Indigenous adults in Canada using the 2022 Canadian Community Health Survey data.

2. Methodology

2.1 Study Design and Data Source

This cross-sectional study used data from the 2022 Canadian Community Health Survey (CCHS), a nationally representative survey conducted by Statistics Canada among individuals aged 12 years and older living in the ten provinces and three territories (Statistics Canada, 2025). For the present analysis, the study population was restricted to Indigenous adults aged 18 years and older, as described below. The survey uses a multistage, stratified sampling design to obtain a representative sample of the Canadian population. Data for individuals living on First Nations reserves and other Indigenous settlements, members of the Canadian Armed Forces, institutionalised populations, and residents of certain remote regions are not included in the CCHS sampling frame. The present study used the 2022 Public Use Microdata File together with the corresponding bootstrap replicate weights provided by Statistics Canada to account for the complex survey design.

2.2 Study Population

The study population consisted of Indigenous adults aged 18 years and older who participated in the 2022 Canadian Community Health Survey. From the original survey sample of 67,079 respondents, 64,843 non-Indigenous participants were excluded, leaving 2,236 respondents who self-identified as Indigenous. A further 190 respondents aged 12 to 17 years were excluded to restrict the analysis to adults, resulting in 2,046 eligible participants. Complete-case analysis was then performed by excluding 414 participants with missing data. The final analytic sample consisted of 1,632 Indigenous adults. After application of the survey sampling weights, this sample represented an estimated 607,893 Indigenous adults living in Canada who were covered by the 2022 Canadian Community Health Survey sampling frame, as shown in Fig. 1 below.

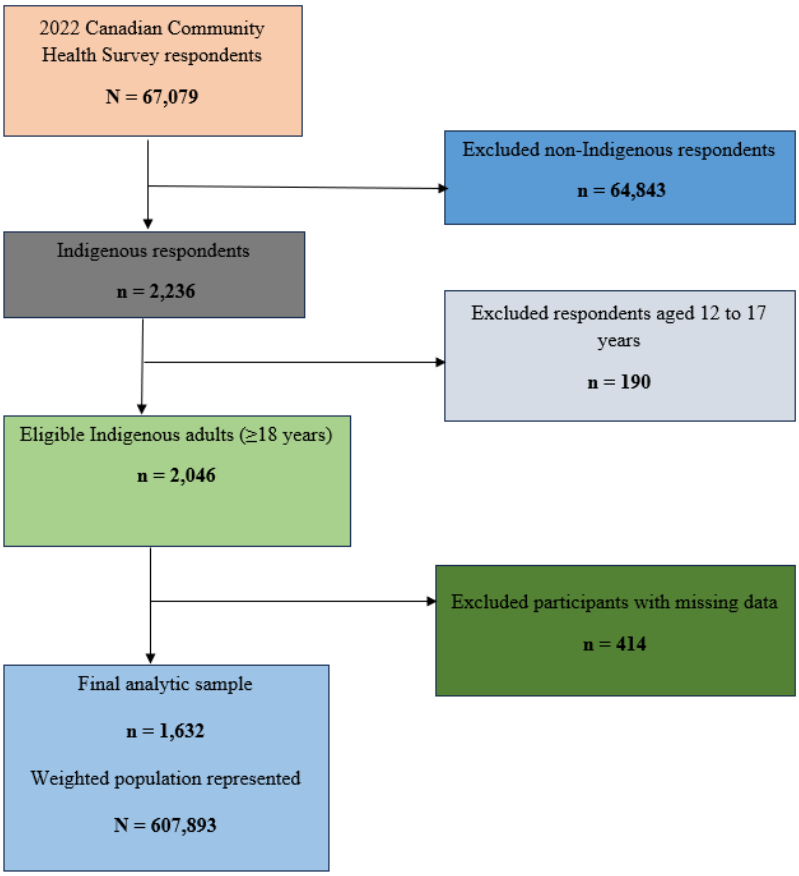


Fig. 1. Flow diagram of participant selection

2.3 Variables and Measures

The outcome variable was self-reported high blood cholesterol, classified as yes or no based on responses to the CCHS chronic conditions module. Independent variables were selected before analysis based on clinical relevance and published evidence regarding factors associated with dyslipidaemia. Demographic variables included age group, sex, and province of residence. Socioeconomic status was represented by the highest level of education attained. Lifestyle factors included body mass index category, smoking status, and alcohol consumption. Clinical factors included self-reported hypertension, diabetes, and cardiovascular disease. Cardiovascular disease, employment status, physical activity, and fruit and vegetable intake were considered during variable selection but were not included because these variables were collected only for subsets of respondents in the 2022 CCHS and contained a large proportion of survey-design valid skips. Their inclusion would have substantially reduced the analytic sample and affected the representativeness of the study population.

2.4 Missing Data

Missing data were assessed before analysis. The proportion of missing values was 5.08% for education, 7.87% for body mass index category, 0.49% for alcohol consumption, 0.34% for high blood cholesterol, 0.49% for hypertension, 3.08% for diabetes, and 5.87% for smoking status. Variables with no missing data included age group, sex, province of residence, Indigenous identity, and survey weights. Complete case analysis was used because the proportion of missing data for the variables included in the final model was low, with the highest level of missingness observed for body mass index category at 7.87%.

2.5 Statistical Analysis

In this study, descriptive statistics were used to summarise participant characteristics. Continuous variables were summarised using survey-weighted means and standard errors, whereas categorical variables were summarised using weighted frequencies and percentages. Group comparisons by the outcome variable were conducted using survey-adjusted t-tests for continuous variables and survey-adjusted chi-square tests for categorical variables. Because variance estimation in the CCHS Public Use Microdata File relies on bootstrap replicate weights, descriptive cross-tabulations were interpreted as descriptive summaries rather than formal hypothesis tests where standard design-based inference was not supported. Multivariable survey-weighted logistic regression was used to identify factors independently associated with high blood cholesterol. The analysis accounted for the complex survey design through the application of sampling weights and bootstrap replicate weights supplied by Statistics Canada. Multicollinearity among the independent variables was assessed using the variance inflation factor (VIF). VIF values ranged from 1.04 to 4.68, with a mean VIF of 1.85, indicating no evidence of problematic multicollinearity. All analyses were performed using Stata version 18 (StataCorp, 2023).

3. Results

Table 1 presents the weighted characteristics of Indigenous adults according to high blood cholesterol status.

Table 1. Characteristics of indigenous adults according to high blood cholesterol status, CCHS 2022

Variable	No High Blood Cholesterol (N = 525,862)	High Blood Cholesterol (N = 82,031)
Sex, n (%)		
Male	268,243 (51.0%)	41,460 (50.5%)
Female	257,619 (49.0%)	40,571 (49.5%)
Age group, n (%)		
18 to 34 years	245,422 (46.7%)	10,290 (12.5%)
35 to 49 years	146,859 (27.9%)	13,449 (16.4%)
50 to 64 years	90,552 (17.2%)	36,880 (45.0%)
65 years and older	43,029 (8.2%)	21,412 (26.1%)
Education, n (%)		
Less than secondary	40,650 (7.7%)	7,620 (9.3%)
Secondary	124,108 (23.6%)	22,021 (26.8%)
Postsecondary	361,104 (68.7%)	52,391 (63.9%)
Province group, n (%)		
Atlantic Canada	22,565 (4.2%)	7,048 (8.6%)
Quebec	44,026 (8.4%)	4,799 (5.9%)
Ontario	116,829 (22.2%)	16,428 (20.0%)
Prairies	106,640 (20.3%)	12,068 (14.7%)
Alberta	100,210 (19.1%)	24,452 (29.8%)
British Columbia	101,463 (19.3%)	12,404 (15.1%)
Territories	34,129 (6.5%)	4,832 (5.9%)
BMI category, n (%)		
Underweight/Normal weight	148,872 (28.3%)	14,691 (17.9%)
Overweight/Obese	376,990 (71.7%)	67,340 (82.1%)

Variable	No High Blood Cholesterol (N = 525,862)	High Blood Cholesterol (N = 82,031)
Smoking status, n (%)		
Current smoker	129,347 (24.6%)	22,245 (27.1%)
Former smoker	112,331 (21.4%)	29,509 (36.0%)
Never smoker	284,184 (54.0%)	30,277 (36.9%)
Alcohol consumption, n (%)		
Regular drinker	319,761 (60.8%)	45,035 (54.9%)
Occasional drinker	108,597 (20.7%)	12,371 (15.1%)
Non-drinker	97,504 (18.5%)	24,626 (30.0%)
Diabetes, n (%)		
No	507,984 (96.6%)	54,977 (67.0%)
Yes	17,878 (3.4%)	27,054 (33.0%)
Hypertension, n (%)		
No	456,929 (86.9%)	44,820 (54.6%)
Yes	68,933 (13.1%)	37,211 (45.4%)

BMI, body mass index; CCHS, Canadian Community Health Survey. Values are weighted frequencies and weighted column percentages. Formal statistical comparisons were not emphasized because variance estimation in the CCHS Public Use Microdata File is based on bootstrap replicate weights, which do not support conventional inference from standard descriptive cross-tabulations. All analyses were performed using Stata version 18 (Tobe et al. 2019)

In this study, the analytic sample represented an estimated 607,893 Indigenous adults in Canada, including 525,862 (86.5%) without high blood cholesterol and 82,031 (13.5%) with high blood cholesterol. Participants with high blood cholesterol were generally older than those without the condition. Nearly one half of participants with high blood cholesterol were aged 50 to 64 years, and about one quarter were aged 65 years and older, whereas almost one half of those without high blood cholesterol were aged 18 to 34 years (245,422; 46.7%). A greater proportion of participants with high blood cholesterol were classified as overweight or obese, had diabetes, and had hypertension. Former smoking and non-drinking were also more common among participants with high blood cholesterol than among those without the condition. The distribution of education, sex, and province group was broadly comparable between the two groups, although some regional differences were observed.

Table 2 presents the results of the survey-weighted multivariable logistic regression analysis examining factors associated with high blood cholesterol among Indigenous adults.

After adjustment for all study variables, age, diabetes, hypertension, and province group remained associated with high blood cholesterol. Compared with participants aged 18 to 34 years, those aged 50 to 64 years had more than five times the odds of reporting high blood cholesterol, adjusted OR 5.16, 95% CI 1.88 to 14.16, $p = 0.001$. Participants aged 65 years and older showed a similar association, adjusted OR 5.80, 95% CI 2.05 to 16.42, $p = 0.001$. Participants with diabetes had substantially higher odds of high blood cholesterol than those without diabetes, adjusted OR 7.06, 95% CI 2.77 to 17.99, $p < 0.001$. Hypertension was also associated with higher odds of high blood cholesterol, adjusted OR 2.36, 95% CI 1.18 to 4.70, $p = 0.015$. Compared with Atlantic Canada, lower odds of high blood cholesterol were observed among participants living in Quebec and the Prairie provinces. No statistically significant associations were found for sex, education, BMI category, smoking status, alcohol consumption, Ontario, Alberta, British Columbia, or the Territories.

Fig. 2 illustrates the weighted prevalence of high blood cholesterol across age groups.

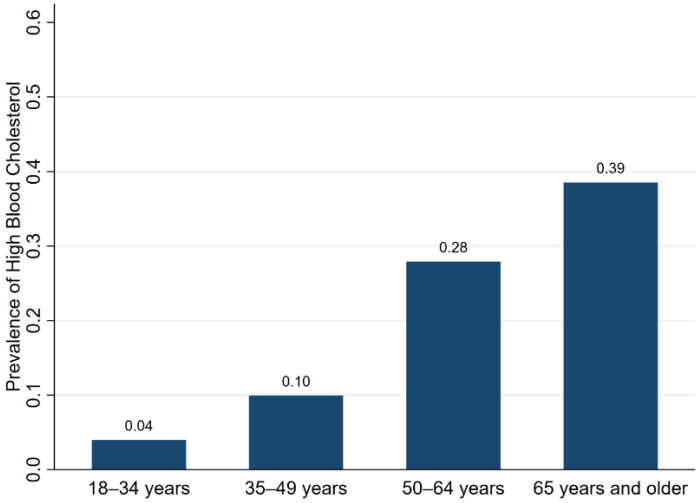


Fig. 2. Weighted prevalence of high blood cholesterol among Indigenous adults by age group, CCHS 2022

Table 1. Survey weighted multivariable logistic regression analysis of factors associated with high blood cholesterol among Indigenous adults in Canada, CCHS 2022

Variable	Adjusted OR	95% CI	p-value
Sex			
Female vs Male	0.94	0.52 to 1.70	0.849
Age group			
35 to 49 years vs 18 to 34 years	1.50	0.58 to 3.83	0.401
50 to 64 years vs 18 to 34 years	5.16	1.88 to 14.16	0.001*
65 years and older vs 18 to 34 years	5.80	2.05 to 16.42	0.001*
Education			
Secondary graduate vs Less than secondary	1.13	0.36 to 3.50	0.832
Post-secondary vs Less than secondary	1.42	0.47 to 4.24	0.533
BMI category			
Overweight/Obese vs Underweight/Normal weight	0.98	0.45 to 2.16	0.965
Smoking status			
Former smoker vs Current smoker	0.94	0.40 to 2.20	0.880
Never smoker vs Current smoker	0.68	0.30 to 1.54	0.352
Alcohol consumption			
Occasional drinker vs Regular drinker	0.95	0.47 to 1.90	0.886
Non-drinker vs Regular drinker	0.87	0.43 to 1.76	0.693
Diabetes			
Yes vs No	7.06	2.77 to 17.99	<0.001*
Hypertension			
Yes vs No	2.36	1.18 to 4.70	0.015*
Province group			
Quebec vs Atlantic Canada	0.25	0.07 to 0.98	0.046*
Ontario vs Atlantic Canada	0.34	0.12 to 1.01	0.053
Prairies vs Atlantic Canada	0.28	0.10 to 0.81	0.018*
Alberta vs Atlantic Canada	0.76	0.30 to 1.89	0.553
British Columbia vs Atlantic Canada	0.47	0.18 to 1.24	0.128
Territories vs Atlantic Canada	0.49	0.21 to 1.16	0.103

OR, odds ratio; CI, confidence interval; BMI, body mass index. Adjusted odds ratios were estimated using survey weighted multivariable logistic regression with bootstrap replicate weights to account for the complex sampling design of the 2022 Canadian Community Health Survey. The model was adjusted for sex, age group, education, BMI category, smoking status, alcohol consumption, diabetes, hypertension, and province group. Reference categories were male, 18 to 34 years, less than secondary education, underweight/normal weight, current smoker, regular drinker, no diabetes, no hypertension, and Atlantic Canada. An asterisk (*) indicates statistical significance at $p < 0.05$. All analyses were performed using Stata version 18 (Tobe et al. 2019)

The prevalence of high blood cholesterol increased steadily with age. Approximately 4% of adults aged 18 to 34 years reported high blood cholesterol, compared with 10% among those aged 35 to 49 years. The prevalence increased to 28% among adults aged 50 to 64 years and reached 39% among those aged 65 years and older. This pattern is consistent with the findings from the multivariable analysis, in which older age groups had higher adjusted odds of high blood cholesterol than younger adults.

4. Discussion

This study examined factors associated with high blood cholesterol among Indigenous adults in Canada using data from the 2022 Canadian Community Health Survey. Increasing age, diabetes, hypertension, and province of residence were independently associated with high blood cholesterol after adjustment for demographic and lifestyle characteristics. Adults aged 50 years and older had substantially higher odds of reporting high blood cholesterol than younger adults. Diabetes showed the strongest association, followed by hypertension. Participants living in Quebec and the Prairie provinces had lower odds of high blood cholesterol than those living in Atlantic Canada. No significant associations were observed for sex, education, body mass index, smoking status, or alcohol consumption after adjustment. These findings support previous evidence showing that cardiometabolic conditions frequently coexist and that Indigenous populations continue to experience a substantial burden of chronic diseases that contribute to cardiovascular risk (Kelly et al., 2019; McCallum et al., 2023; Shahunja et al., 2024).

The strong association between older age and high blood cholesterol is consistent with the natural progression of lipid metabolism across the life course. Advancing age is accompanied by changes in lipid handling, vascular function, and cumulative exposure to cardiovascular risk factors, increasing the likelihood of dyslipidaemia (Zhao et al., 2024). The higher odds of high blood cholesterol among participants with diabetes are also consistent with current evidence showing that abnormalities in glucose metabolism and lipid metabolism commonly occur together (Hu et al., 2022; Ray et al., 2019). Insulin resistance may contribute to elevated concentrations of atherogenic lipoproteins, while persistent hyperglycaemia can further disrupt lipid regulation, increasing cardiovascular risk (Ray et al., 2019). Hypertension was also associated with high blood cholesterol in this study. These conditions frequently coexist because they share several underlying metabolic and behavioural risk factors, including obesity, unhealthy dietary patterns, and reduced physical activity (Hu et al., 2022; Tobe et al., 2019). The coexistence of these conditions is clinically important because cardiovascular risk increases when multiple cardiometabolic disorders are present simultaneously (Kelly et al., 2019; McCallum et al., 2023; Shahunja et al., 2024).

Current Canadian recommendations emphasise the early identification and management of dyslipidaemia as part of comprehensive cardiovascular risk reduction. The Canadian Cardiovascular Society recommends routine assessment of lipid profiles in adults with established cardiovascular risk factors, including diabetes and hypertension, together with lifestyle modification and lipid-lowering therapy when clinically indicated (Pearson et al., 2021). These recommendations are consistent with the present findings, in which diabetes and hypertension remained independently associated with high blood cholesterol after adjustment. Early recognition of dyslipidaemia is particularly important among Indigenous populations because cardiovascular disease continues to contribute substantially to preventable illness and premature mortality (McCallum et al., 2023). Approaches that integrate screening, risk assessment, culturally appropriate education, and long-term follow-up may help improve cardiovascular health in these communities (McCallum et al., 2023; Tobe et al., 2019).

Several variables that are commonly associated with dyslipidaemia were not statistically significant after multivariable adjustment. Sex, educational attainment, body mass index, smoking status, and alcohol consumption were not associated with high blood cholesterol after adjustment for other characteristics. This pattern suggests that the observed differences may be influenced by the presence of stronger cardiometabolic factors, particularly diabetes and hypertension, which remained independently associated with the outcome. Similar findings have been reported in studies showing that chronic metabolic conditions often have a greater influence on dyslipidaemia than individual lifestyle characteristics once multiple factors are considered simultaneously (Hu et al., 2022; McCallum et al., 2023).

5. Strengths and Limitations of the Study

This study has several strengths. It used a large, nationally representative survey and incorporated survey weights together with bootstrap replicate weights to account for the complex sampling design, allowing the findings to represent Indigenous adults covered by the Canadian Community Health Survey sampling frame.

Several limitations should also be considered. The cross-sectional design does not allow the temporal sequence between exposures and high blood cholesterol to be established. All variables were self-reported and may therefore be affected by recall error or misclassification. Complete-case analysis was used to address missing data, which reduced the analytic sample. Employment status, physical activity, and fruit and vegetable consumption were not included because these variables contained a large proportion of valid skips that would have substantially limited the eligible sample. Cardiovascular disease was also excluded because of extensive missing data. Future longitudinal studies that include objective clinical measurements together with more complete information on lifestyle, social determinants of health, and access to preventive care would strengthen the understanding of factors associated with high blood cholesterol among Indigenous adults in Canada.

6. Conclusion

This cross-sectional analysis of the 2022 Canadian Community Health Survey identified older age, diabetes, hypertension, and province of residence as factors associated with self-reported high blood cholesterol among Indigenous adults represented in the survey. Adults aged 50 years and older had substantially higher adjusted odds of high blood cholesterol than those aged 18–34 years, while diabetes and hypertension were also independently associated with the outcome. Geographic differences were observed, with lower adjusted odds in Quebec and the Prairie provinces compared with Atlantic Canada. In contrast, sex, educational attainment, body mass index, smoking status, and alcohol consumption were not significantly associated after multivariable adjustment. These findings indicate that high blood cholesterol co-occurs with important cardiometabolic conditions and may be usefully considered within integrated cardiovascular risk assessment. Because the study was cross-sectional and based on self-reported data, the findings should be interpreted as associations rather than causal relationships. Further longitudinal research using objective clinical measurements would help clarify these relationships.

Consent

It is not applicable.

Ethical Considerations

The study used de identified publicly available data obtained from the 2022 Canadian Community Health Survey Public Use Microdata File. Individual participants cannot be identified from the dataset. According to institutional policies for secondary analysis of publicly available anonymized data, additional research ethics board approval and informed consent were not required.

Declaration of AI Use

This manuscript was prepared through the intellectual contributions of the author(s) to the study design, data analysis, interpretation of results, and scholarly content. Grammarly and ChatGPT were used solely to assist with grammatical refinement and reference formatting during manuscript revision. The author(s) accept full responsibility for the accuracy, integrity, and final content of the manuscript.

Competing Interests

Authors have declared that no competing interests exist.

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